

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Dot .Net Core and Progressive Web Application Practical

Practical - X

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PART B – Questions

- a. Create a PWA with a Web App Manifest file including: App name Icons Start URL Display mode Implement Push Notifications in a PWA.

Code:

manifest.json

```
{
  "name": "My PWA Application",
  "short_name": "MyPWA",
  "start_url": "/index.html",
  "display": "standalone",
  "background_color": "#ffffff",
  "theme_color": "#0d6efd",
  "icons": [
    {
      "src": "icons/icon1.png",
      "sizes": "192x192",
      "type": "image/png"
    },
    {
      "src": "icons/icon2.png",
      "sizes": "512x512",
      "type": "image/png"
    }
  ]
}
```

Index.html:

```
<!DOCTYPE html>
<html>
<head>
  <meta charset="utf-8" />
  <title>My PWA Application</title>
  <link rel="manifest" href="manifest.json">
  <meta name="theme-color" content="#0d6efd">
</head>
```

```
<body>
  <h1>My Progressive Web App Application</h1>
  <button onclick="showNotification()">Show Notification</button>
  <script src="app.js"></script>
</body>
</html>
```

app.js:

```
if ('serviceWorker' in navigator)
{
  navigator.serviceWorker.register('service-worker.js')
    .then(() => console.log('Service Worker registered successfully'));
}
function showNotification()
{
  Notification.requestPermission().then(permission =>
  {
    if (permission === 'granted')
    {
      navigator.serviceWorker.ready.then(registration =>
      {
        registration.showNotification('PWA Notification', {
          body: 'Push notification Working Successfully!',
          icon: 'icons/icon1.png'
        });
      });
    }
  });
}
```

service-worker.js

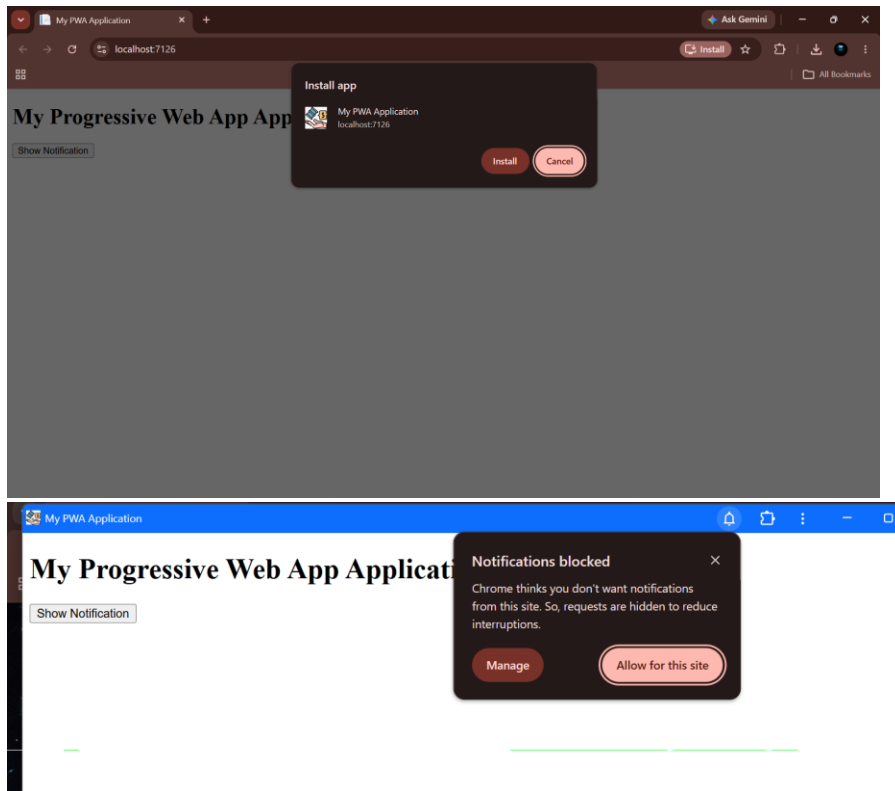
```
self.addEventListener('install', event => {
  console.log('Service Worker: Installed');
});
self.addEventListener('activate', event =>
{
  console.log('Service Worker: Activated');
});
```

program.cs

```
var builder = WebApplication.CreateBuilder(args);
var app = builder.Build();
app.UseDefaultFiles();
app.UseStaticFiles();

app.Run();
```

Output:



- b. Develop an E-Commerce PWA with product listing and offline cart functionality.
Code:

[program.cs](#)

```
var builder = WebApplication.CreateBuilder(args);
var app = builder.Build();
```

```
app.UseDefaultFiles();
app.UseStaticFiles();
```

```
app.Run();
```

manifest.json

```
{
  "name": "ECommerce PWA",
  "short_name": "shopPWA",
  "start_url": "/index.html",
  "display": "standalone",
  "background_color": "#ffffff",
  "theme_color": "#0d6efd",
  "icons": [
    {
      "src": "icons/icon1.png",
      "sizes": "128x128",
      "type": "image/png"
    },
    {
      "src": "icons/icon2.png",
```

```
        "sizes": "512x512",
        "type": "image/png"
    }
  ]
}
```

index.html

```
<!DOCTYPE html>
<html>
<head>
  <meta charset="utf-8" />
  <title>E-Commerce PWA</title>
  <link rel="manifest" href="manifest.json">

  <meta name="theme-color" content="#0d6efd">
  <style>
    body{
      font-family:Arial;
      margin:20px;
    }
    .container {
      display: flex;
      gap: 30px;
    }
    .cart-section{
      width:30%;
      border:1px solid gray;
      padding: 15px;
      min-height: 300px;
    }
    .product-section{
      width:70%;
    }
    .products {
      display: flex;
      flex-wrap: wrap;
      gap: 15px;
    }

    .product {
      border: 1px solid gray;
      padding: 10px;
      width: 200px;
    }

    button {
      padding: 5px 10px;
      cursor: pointer;
    }
  </style>
</head>
<body>
  <h1>E-Commerce PWA</h1>
```

```
<div class="container">

  <!-- Cart Section -->
  <div class="cart-section">
    <h2>Cart Items</h2>
    <ul id="cartList"></ul>
  </div>

  <!-- Product Section -->
  <div class="product-section">

    <h2>Products</h2>

    <div class="products">

      <div class="product">
        <h3>Laptop</h3>
        <p>Price: ₹ 80000</p>
        <button onclick="addToCart('Laptop')">
          Add to Cart
        </button>
      </div>

      <div class="product">
        <h3>Mobile</h3>
        <p>Price: ₹ 25000</p>
        <button onclick="addToCart('Mobile')">
          Add to Cart
        </button>
      </div>

      <div class="product">
        <h3>Headphones</h3>
        <p>Price: ₹ 5000</p>
        <button onclick="addToCart('Headphones')">
          Add to Cart
        </button>
      </div>

    </div>
  </div>

  <div>

    <script src="app.js"></script>

  </div>

</body>
</html>
```

app.js

```
if ('serviceWorker' in navigator) {
  navigator.serviceWorker.register('service-worker.js')
    .then(() => console.log('Service Worker Registered'));
```

```

}

let cart = JSON.parse(localStorage.getItem('cart')) || [];

displayCart();

function addToCart(product) {
  cart.push(product);

  localStorage.setItem('cart', JSON.stringify(cart));

  displayCart();
}

function displayCart() {
  let cartList = document.getElementById('cartList');

  cartList.innerHTML = "";

  cart.forEach(item => {
    let li = document.createElement('li');

    li.textContent = item;

    cartList.appendChild(li);
  });
}

```

server-worker.js

```

const CACHE_NAME = 'ecommerce-cache-v1';

const urlsToCache = [
  '/',
  '/index.html',
  '/app.js',
  '/manifest.json'
];

self.addEventListener('install', event => {
  event.waitUntil(
    caches.open(CACHE_NAME)
      .then(cache => {
        return cache.addAll(urlsToCache);
      })
  );
});

self.addEventListener('fetch', event => {
  event.respondWith(
    caches.match(event.request)
      .then(response => {
        return response || fetch(event.request);
      })
  );
});

```

});

Output:

